

Angel Gabriel Munguia Velazquez

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EDUCATION

Universidad de las Américas Puebla (UDLAP) <i>B.S. in Physics, Honors Program (GPA: 3.9/4.0)</i> Honors: Full Academic Excellence STEM Scholarship (100% Tuition) – <i>Awarded to top-tier STEM talent.</i> Relevant Coursework: Ordinary Differential Equations, Mathematical Analysis I, Numerical Analysis, Matrix Theory.	Puebla, Mexico <i>Expected May 2028</i>
Bachillerato 5 de Mayo - BUAP <i>International Baccalaureate (IB) Diploma Programme</i> Honors: Graduated with 34 IB points – <i>Rigorous pre-university program emphasizing scientific research.</i>	Puebla, Mexico <i>May 2024</i>

RESEARCH EXPERIENCE

Universidad de las Américas Puebla (UDLAP) <i>Undergraduate Researcher – Honors Program</i> Investigating the formation of high-redshift galaxies utilizing Machine Learning to refine predictive models for classification.	Puebla, Mexico <i>Aug 2025 – Present</i>
Mexican Physical Society (SMF) <i>Poster Presenter – 67th National Physics Congress</i> Presented experimental research characterizing magneto-optical properties of organic compounds (Faraday Effect) via a self-designed experimental setup.	Chihuahua, Mexico <i>Oct 2024</i>

ENGINEERING & TECHNICAL PROJECTS

Ibero-American Infomatrix Final <i>AI Developer – Gold Medalist & Infomatrix Argentina Qualifier</i> - Developed an AI visual assistant for the visually impaired, integrating Multimodal LLMs with YOLO object detection. - Engineered a low-latency video streaming pipeline over Wi-Fi utilizing an Arduino UNO Q to transmit real-time webcam feeds.	San Luis Potosi, Mexico <i>Mar 2026</i>
Torneo Mexicano de Robótica (TMR) <i>Lead of Electronics and Control – Beach Cleaning Robots</i> Engineered hardware and control systems for beach-cleaning robots, spanning from Arduino-based logic (2025) to custom PCBs with ESP32 microcontrollers (2026).	Mexico <i>2025 – 2026</i>
NASA Space Apps Challenge <i>AI Developer & iOS App Developer (Team Collaboration)</i> 2025 (Air Quality): Engineered the predictive core (LSTM, XGBoost) and integrated LLMs for a health platform. 2024 (Exoplanet Mapping): Built a Swift iOS interface for real-time star mapping utilizing NASA datasets.	Puebla, Mexico <i>2024 – 2025</i>
iOS Development Lab & Apple Swift Challenge <i>iOS App Developer – National Finalist</i> Built native iOS apps in Swift, applying memory management and UI/UX principles for end-to-end deployment.	Puebla, Mexico <i>May 2025</i>

ACADEMIC LEADERSHIP & OUTREACH

UDLAP Official STEM Representative Team <i>Inaugural STEM Scholar & Lead Instructor</i> Summer 2026: Instructed “A very brief Introduction to Quantum Computing”. Summer 2025: Instructed “Entre historias y juegos STEM”, a science communication workshop.	Puebla, Mexico <i>May 2025 – Present</i>
Puebla Planetarium Dome / Feria de Puebla 2026 <i>Guest Speaker – “Cosmic Shortcuts”</i> Delivered an immersive astrophysics lecture inside the main dome for large public audiences, driving STEM engagement.	Puebla, Mexico <i>May 2026</i>
“Noche de las Estrellas” Outreach Program <i>Workshop Designer & Presenter</i> Facilitated public science workshops scaling from Electromagnetism (2024) to Machine Learning in Astrophysics (2026).	Puebla, Mexico <i>2024 – 2026</i>
Technical Conference Organization <i>Event Staff</i> Facilitated technical operations for the International Micro Air Vehicle Conference (IMAV) and MEXCIR.	Mexico <i>2025</i>

HONORS & AWARDS

Gold Medalist & Argentina 2026 Qualifier , Ibero-American Infomatrix Final	2026
National Finalist (5th Place Regional) , iOS Development Lab Hackathon	2025
National Competitor , Mexican National Physics Olympiad	2024
4th Place (State Level) , Mexican Physics Olympiad	2023

TECHNICAL SKILLS & CERTIFICATIONS

Comp. Physics, AI & Quantum: Python (NumPy, PyTorch, TensorFlow, scikit-learn), C, Matlab, YOLO, LLMs, LSTM, XGBoost.
Hardware: Custom PCB Design, Microcontrollers and microcomputers (ESP32, Arduino, Raspberry Pi).
Certifications: Introduction to Deep Learning (CU Boulder), Numerical Methods for Engineers (HKUST).
Tools & Languages: Quarto, L ^A T _E X, Git/GitHub English (TOEFL ITP: 587).